

Prague University of Economics and Business

Faculty of Informatics and Statistics



**POWER DEMAND FORECASTING USING
MACHINE LEARNING APPROACHES**
BACHELOR THESIS

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Abstract

This thesis explores the application of various machine learning models to power demand forecasting, a crucial component in the efficient management of energy resources. The main goal of this thesis is to introduce the machine learning models used for forecasting power demand and compare those methods. Additional goals supporting the main goal are contributing to the optimization of the German energy grid, demonstration of the practical application using real data, and preparation for future research. Those goals were fulfilled by using literature and the author's experience. Python programming language was used to implement the whole practical part. The outcome of this thesis is a concise comparison of different models, such as linear regression, XGBoost and long short-term memory neural network.

Keywords

energy grid, electricity demand forecasting, linear regression, LSTM, machine learning, XGBoost.

Abstrakt

Tato práce se zabývá aplikací různých modelů strojového učení na předpovídání poptávky po elektřině, což je klíčová součást efektivního řízení energetických zdrojů. Hlavním cílem této práce je představit modely strojového učení používané pro předpovídání poptávky po elektřině a tyto metody porovnat. Dalšími cíli podporujícími hlavní cíl jsou přispění k optimalizaci německé energetické sítě, ukázka praktického použití na reálných datech a příprava na budoucí výzkum. Tyto cíle byly naplněny s využitím literatury a zkušeností autora. K realizaci celé praktické části byl použit programovací jazyk Python. Výsledkem této práce je stručné srovnání různých modelů, jako je lineární regrese, XGBoost a long short-term memory neuronová síť.

Klíčová slova

energetická síť, předpověď poptávky po elektřině, lineární regrese, LSTM, strojové učení, XGBoost.

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List of Abbreviations

BT	Bachelor Thesis
DT	Decision Tree
ENTSO-E	The European Network of Transmission System Operators for Electricity
FIS	Faculty of Informatics and Statistics
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MSE	Mean Squared Error
RF	Random Forest
RMSE	Root Mean Squared Error
XGB	XGBoost

Introduction

Effective power demand forecasting is crucial in an era of increasing reliance on renewable energy sources and the growing impact of environmental factors (ex. temperature, wind, etc.). As the energy sector has a trend to transition away from fossil fuels, the ability to accurately predict power demand becomes increasingly important for stability of energy grid and economic efficiency. Traditional forecasting methods face challenges in accurately integrating these dynamic elements, often resulting in inoptimal energy management.

This BT will explore the application of different ML approaches to showcase the performance differences in various methods. It will evaluate a range of models, including neural networks, regression methods and decision trees, and qualify their effectiveness in handling the unpredictable nature of renewable energy and environmental impact.

The comparison must include various characteristics, such as accuracy, that would be the most important metric. Also, it should include the difference in efficiency and ease of interpretation. That would involve not only quantitative analysis through statistical measurements but also qualitative analysis regarding the interpretation of results.

The main goal of this thesis is to explore existing power demand forecasting techniques, particularly focusing on using meteorological data such as temperature, wind, and humidity. The primary objective is to compare the most popular ML methods and achieve optimal performance. Additional aims include contributing to the optimization of the energy grid, which supports the current shift towards more renewable energy sources. This study will also highlight the challenges encountered during the implementation of various methods and offer potential solutions, thereby providing a practical guide for addressing these issues.

The expected outcome of this BT would be the comparison of different algorithms, considering factors such as the complexity of implementation, training and preparation time required, and interpretability of the results. By highlighting the strengths and weaknesses of each algorithm, this comparison will contribute to the ongoing optimization of power demand forecasting algorithms.

Goals

This BT reviews existing power demand forecasting techniques and emphasizes their interaction with temperatures and renewable energy data. The main goal of this thesis is to introduce the ML models used for forecasting power demand and compare those methods. This should bring clarity to the potential of different methods used by the people in this field. We will perform a comparative analysis of traditional statistical methods and ML techniques to determine which models provide the most accurate and reliable forecast.

Additional goals that this BT will try to achieve are:

- contribute to the optimization of the grid, ultimately supporting the transition towards a more sustainable energy system
- demonstrate the practical application of the selected methods by using real-world data, explaining the challenges and their solutions made along the implementation process
- prepare the basis for future research directions based on the outcome of this BT, suggesting further improvements and sustainability of energy grids

Limitations

Certain limitations constrain this BT. The focus is divided between univariate and multivariate time series forecasting, using historical load data from ENTSO-E platform and weather data taken from OpenWeather API as inputs. While this thesis acknowledges the different aspects of predicting energy data, incorporating variables such as demographic or economic indicators will not be implemented here, as it would drastically increase the difficulty for a small boost in performance.

Furthermore, the analysis is limited to publicly available data and computational resources available at hand.

All the necessary permissions to use the data have been obtained from the ENTSO-E and OpenWeather API to facilitate this thesis.

Methods

The methodology consists of a detailed analysis of Germany's energy grid. Data is collected from the ENTSO-E Transparency Platform and OpenWeather API, preprocessed using Python's Pandas library, and analyzed with a suite of ML algorithms.

The BT involves a theoretical introduction to the German and European power grids, an explanation of the most common techniques used for power demand prediction, and a summary of the practical part that consists of finding and loading the data sources, preprocessing it, making the preparations for each model, and finally implementing the models. The final step would be empirical testing to evaluate the performance of the forecasting methods.

Structure

The thesis is structured into two primary sections. The first section establishes the theory behind the energy infrastructure of the German and European energy grids, introduces the ML methods used in forecasting demand, and outlines the evaluation metrics used to compare the results.

The second section dives deeper into the practical application of explained terminology, exploring the tools and technologies employed, and goes into the process of collecting the data and preparing it for the further implementation of ML models.

Expected results

The expected outcomes of this BT include a comparative analysis of traditional forecasting methods and ML approaches, with a demonstration of their performance in capturing the complexities of power demand. It should outline the difference between the most popular approaches and aim to explain the outcomes.

It will also aim to offer some insights that can improve decision-making within the energy sector, contributing to more efficient and reliable grid management.

1 Theoretical part

The theoretical part of this thesis will explain the foundational information needed to understand how the energy grid works within the context of Germany's and Europe's energy grids. This section starts by examining the structure of the Energy Transition (Energiewende) for Germany's energy sector and the advancements of smart grid technologies. The influence of those factors on power demand forecasting is analyzed to understand their impact on predictive accuracy and reliability.

After this, it broadens to the European Energy Grid to explore the integration of national grids and potential future developments. This then includes an explanation of the interconnected nature of the European country's grids and what it means for energy policy and planning.

Furthermore, this section goes into ML methods that are used for power demand forecasting. It compares traditional statistical techniques, such as linear regression, against more modern approaches like decision trees and neural networks, highlighting the advantages and limitations of each.

Presentation of all those elements is crucial for a comprehensive understanding of current energy systems, the role of forecasting in these systems, and how ML can enhance the prediction process.

1.1 Germany's Energy Grid

Germany's energy grid is a complex and highly integrated network designed to distribute electricity from producers to consumers across the country. It is part of the larger European interconnected system, playing a key role in regional energy security and market dynamics.

In this chapter, we will look closer at Germany's energy grid structure and what fuel types create the country's energy stack. Additionally, it will describe the Energy Transition initiative called Energiewende, which aims to reduce the use of nuclear and fossil fuels for a more sustainable future.

Further, it will focus on other technological innovations that will contribute to a more sustainable grid in the future, like smart grids or energy storage systems, that should help in the future development of renewable energy sources. Since the reliability of those sources is low, they should help solve the issue of inconsistent energy generation.

1.1.1 Grid structure

Germany's energy generation portfolio is diverse, consisting of coal, natural gas, nuclear power, and a significant and growing proportion of renewable energy sources like wind,

solar, and biomass. This diverse energy stack is an important part of Germany's energy policy, aiming for sustainability and reduced carbon emissions.

As we can see in the Figure 1, Germany's energy grid structure in 2024 already shows a strong commitment to renewable energy, with solar and wind power leading in installed capacity. Solar energy leads as the dominant source, showing a significant investment in this sustainable technology. Wind energy, both onshore and offshore, also makes up a large portion of the energy mix, while fossil fuels like coal, oil, and gas show smaller capacities. This indicates a strategic move towards renewable resources, aligning with Germany's Energiewende, that would be explained more in the next section. Smaller contributions from hydro and biomass complete the diverse energy portfolio, supporting the overall stability and sustainability of the grid.

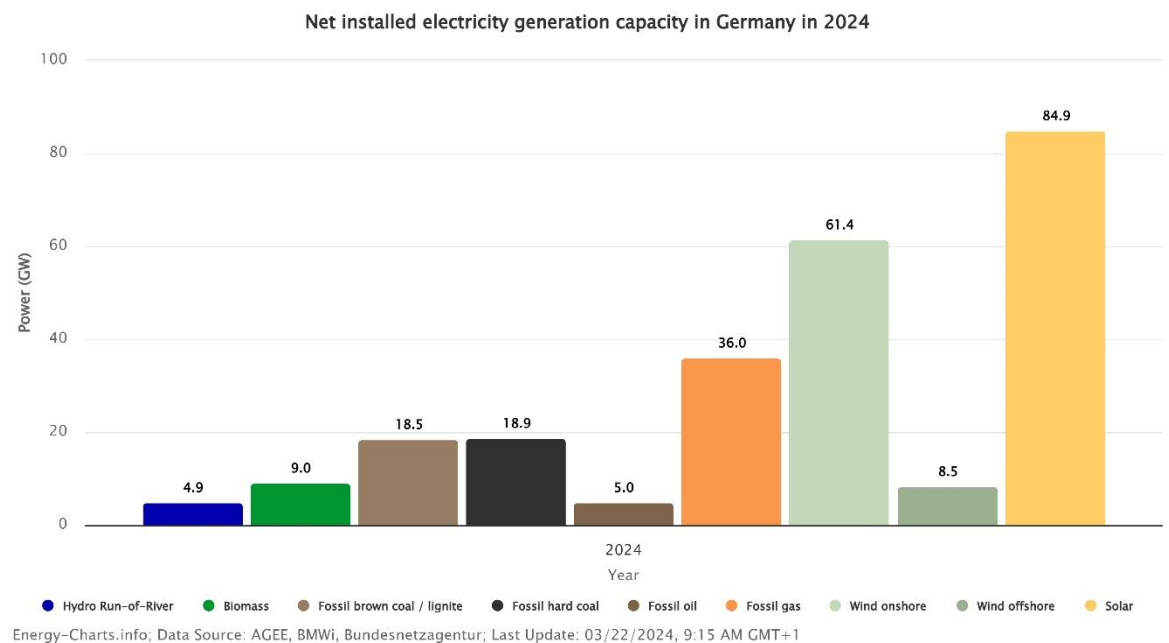


Figure 1 Germany energy generation portfolio (Source: Energy charts 2024)

The grid consists of high-voltage transmission lines that carry electricity over long distances and lower voltage distribution networks that deliver power to end-users. The transmission grid is operated by four major Transmission System Operators (TSOs): TenneT, Amprion, 50Hertz, and TransnetBW (International Trade Administration 2023).

The grid operates under strict regulations set by the Bundesnetzagentur (Federal Network Agency), which is responsible for grid access and monitoring electricity market integration.

German greenhouse gas emissions by sector 1990-2022 and mandated emission budgets 2020-2030

Data: UBA (2023) / Climate Action Law (2021).

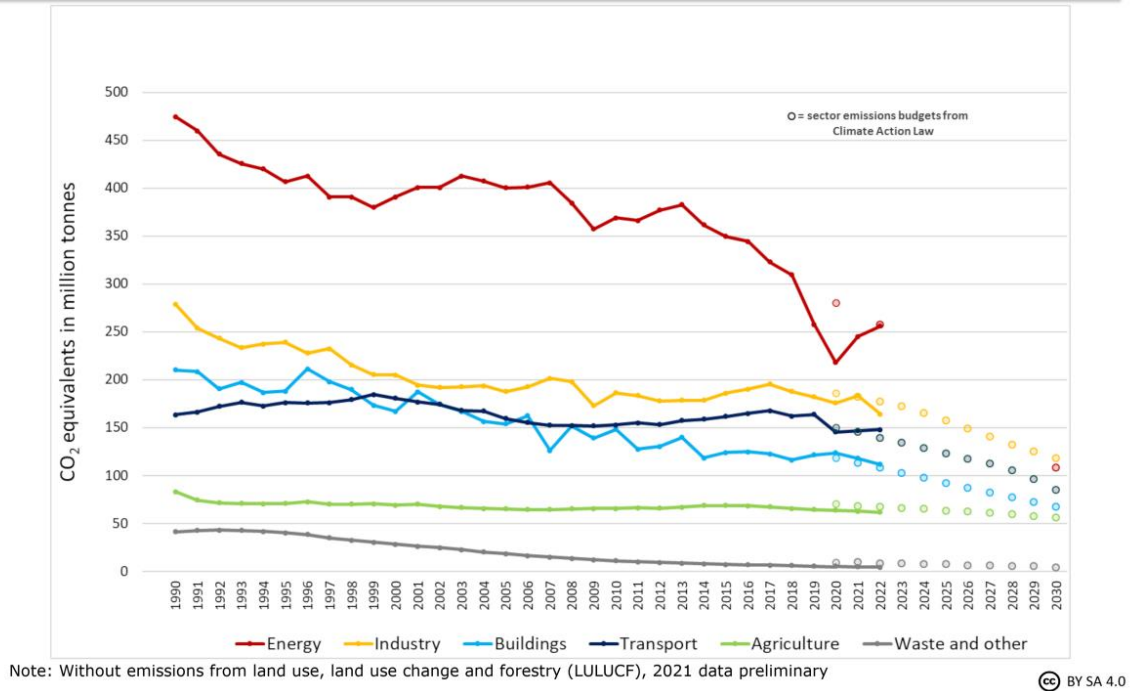


Figure 3 German greenhouse gas emissions (Source: Appunn, Eriksen, Wettengel 2024)

1.1.3 Technological innovations and Smart Grid development

To accommodate a high percentage of renewables, Germany is investing in smart grid technology and energy storage. Smart grids help manage electricity demand and supply more efficiently and reliably, while energy storage systems are important for balancing load and providing power despite inconsistent energy generation from renewables (International Trade Administration 2023).

Summary

Understanding Germany's energy grid and its policies on that matter is a crucial factor for understanding the bigger perspective of how the demand is working there. It helps to paint a fuller picture of the long-term trends and helps develop more accurate forecasting models, which becomes harder as the grid incorporates more and more renewable energy sources. The ability to predict demand accurately helps in planning the necessary energy generation and storage, minimizing the cost, and maximizing the reliability of the energy supply.

1.2 The European Energy Grid

This section will dive deeper into the European energy grid, which represents a complex web of interconnected national grids spread around Europe. Those interconnections

provide a possibility for energy trade and distribution across national borders, improving reliability and energy security for member countries.

This section examines the European grid structure, focusing on the interconnections between countries and the challenges it faces. It will discuss the solutions the EU is implementing, and the direction Europe is taking to manage power across the continent.

Understanding those factors is crucial for grasping how regulations and policies impact power demand. It would also contribute to understanding how the unpredictable nature of renewable energy could affect the quality of predictive models and how it could be counted in the model.

1.2.1 Structure and integration

This part will focus on the structure of the European energy grid, explaining the integration of the network across national boundaries and the operational coordination that facilitates energy distribution in Europe. The importance of ENTSO-E as a data provider should also be mentioned, as it plays a key role in balancing power across the regions, allows for more efficient regulation and provides key information for the energy sector.

The European grid is characterized by numerous cross-border interconnections that allow for the movement of electricity across the continent. These connections are crucial for balancing supply and demand across regions, allowing countries with surplus energy to export to those facing deficits (Cremona 2023).

The European Network of Transmission System Operators for Electricity (ENTSO-E) is an online platform that provides data from the European electricity system and plays a pivotal role in coordinating the operation of the European electricity market. It ensures that technical and regulatory standards are maintained and that the grid operates efficiently and reliably (Hirth, Mühlenpfordt, Bulkeley 2018).

1.2.2 Challenges

There are many challenges that are present in European Energy Grid, based on 2017 journal article called “European smart grid prospects, policies, and challenges” from IqtiyaniIlham and Hasanuzzaman (IqtiyaniIlham, Hasanuzzaman 2017) several key issues are:

- **Regulatory complexity**

The integration of multiple national grids involves navigating a complex landscape of regulations and market mechanisms. Each country has its own regulations, and bringing such different regulatory frameworks is challenging.

- **Infrastructure needs**

As been noted in the section about challenges of Energiewende, the demand for renewable energy increases and the European Union pushes for a greener energy policy, significant

investment in grid infrastructure is needed to handle the increased load and variability introduced by renewables like wind and solar power.

- **Energy security**

While interconnectivity enhances energy security by diversifying sources, it also introduces vulnerabilities, as grid stability can be affected by problems in interconnected areas. Additionally, geopolitical tensions can impact the security of energy supplies, particularly for countries heavily reliant on imports.

1.2.3 Future directions

The EU's Energy Union Strategy aims to provide secure, sustainable, competitive, and affordable energy across the continent. A key component of this strategy is the further integration of the energy markets, ensuring that all Europeans can benefit from secure and affordable energy (Far, Youngs 2015).

Europe is increasingly focusing on digitalizing its energy grid, incorporating smart grid technologies to improve efficiency and flexibility. This includes developing smart meters, demand response technologies, and better grid management technologies to handle the fluctuations associated with high levels of renewable energy (Palensky, Kupzog 2013).

Another direction that Europe is taking is the global effort toward decarbonization. Future grid development will heavily involve integrating more renewable energy sources, necessitating innovative solutions in energy storage, grid management, and interconnectivity to maintain reliability and stability.

1.2.4 Influence on power demand forecasting

The interconnected nature of the European grid means that forecasting must consider not only national demand but also the dynamics of cross-border energy flows. Forecasters need to account for various factors, including foreign market influences, exchange rates, and international policy changes.

Next way how the mentioned factors influence the power demand forecasting is high volatility of renewable energy sources. Forecasting models need to become more sophisticated, integrating meteorological data and adapting to rapid changes in energy availability.

1.3 Machine Learning methods

This chapter will focus on the ML methods that will be used in this BT. It will thoroughly explain various models, outlining how they work and their advantages and disadvantages, with a focus on their influence on power demand forecasting. It will go through traditional methods such as LR to more modern techniques like XGB and Neural Networks, particularly using the LSTM model.

1.3.1 Regression methods

Linear regression is a fundamental statistical method for predicting how one variable change based on other variables. This method requires the relationship between the variables to be linear. It is straightforward and interpretable, which makes it a popular choice for initial analysis and forecasting.

However, it is effective only in scenarios where the relationships between variables are indeed linear and do not involve complex or non-linear patterns. The formula for linear regression is:

$$y = \alpha + \beta x \quad (1)$$

In this formula, y is the variable we are trying to predict, α is where the line starts, β shows how steep the line is, and x is the variable we are using to predict Y .

Linear regression assumptions

Based on the blog post from Bobbitt Zach about the assumptions of LR, we can outline 4 main assumptions (Bobbitt 2020):

1. **Linearity:** The relationship between the independent and dependent variables is supposed to be linear.
2. **Independence:** The residuals do not depend on each other.
3. **Homoscedasticity:** The spread of residuals is consistent no matter the value of the independent variable.
4. **Normality:** The residuals of the model are normally distributed.

Strengths and limitations

According to Naresh Kumar the advantages and disadvantages of Linear Regression in ML (Kumar 2019) are as follows:

Strengths

- It is simple to understand and interpret.
- It is flexible, capable of modeling different kinds of relationships, not just straight lines. With some adjustments, even curved relationships can be made to look straight for analysis.

Limitations

- It does not handle outliers well; unusual data points can skew the analysis. Cleaning the data can help mitigate this.
- Avoiding overfitting and underfitting is necessary. Overfitting happens when the model fits the training data too closely and fails with new data. Underfitting is when the model is too simple to capture the underlying trend.

1.3.2 Decision Trees methods

Decision Trees are a non-parametric supervised learning method that can be utilized for both classification and regression tasks. They stand out as powerful tools for statistical analysis due to their ability to efficiently manage large datasets during classification. The core concept of Decision Trees involves a process of recursive partitioning, where the dataset is split into increasingly smaller groups. These groups are more homogeneous regarding the outcome variable within each group and more heterogeneous between the groups. This method, developed by Breiman, Friedman, Stone, and Olshen in 1984 (Breiman, Friedman, Olshen, Stone 1984), simplifies the analysis of how dependent variables relate to independent variables.

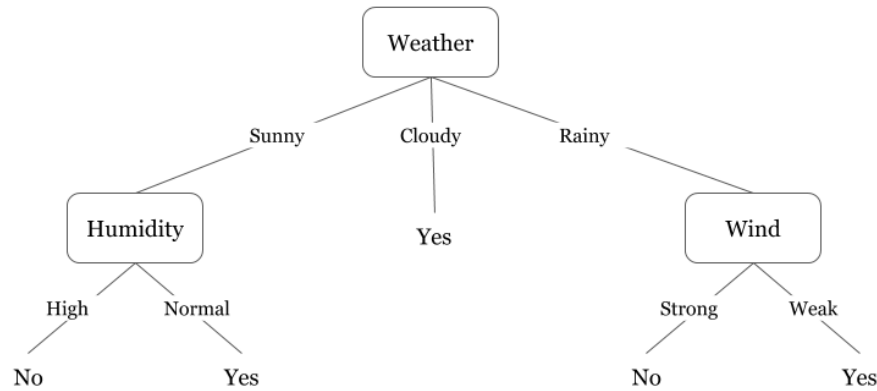


Figure 4 Example of a decision tree (Source: Kumar 2020)

The creation of a DT is notably straightforward. It assesses various conditions and aligns the outcomes in a simple *IF(Condition) THEN(Result)* manner. This method divides the dataset into two or more segments, with each segment corresponding to a specific condition (Kapustina 2021).

DT consists of 3 types of components, which are the root node, decision node, and leaf node. The root node is a node where the tree starts its division, the decision node is a node that is placed after root node, and the leaf node is a final node in a branch where no further splitting is possible (Saini 2022).

Feature selection

Selecting the root node in decision tree models involves determining the feature that best splits the data, for which the Gini index is used. This metric assists in evaluating the “impurity” of a split, which refers to how homogeneous the split is in terms of the target variable. The Gini index measures the likelihood of incorrect classification of an element and is mathematically written as:

$$Gini\ index = 1 - [(P_+)^2 + (P_-)^2] \quad (2)$$

Where P_+ is the probability of a positive class and P_- is the probability of a negative class. The lower the Gini index is, the lower the impurity, which is what we are looking for (Saini 2022).

Random Forest

Random Forest is a popular decision tree method. Its most significant advantages are accuracy and robustness, making it a popular choice for various ML applications. RF is a technique that consists of many decision trees. Instead of depending on one tree, it creates multiple trees and, based on the majority votes of predictions, predicts the final output, which can be better seen in Figure 5 (Saini 2022).

Gradient Boosting

Boosting is an ML technique that can be used for regression and classification problems. In boosting the trees are built sequentially, meaning that each subsequent tree would learn from the errors of the previous tree, which can be visually seen in Figure 5. This approach not only helps increase accuracy, but it is also computationally efficient and scalable. Opposed to the regular parallel method called bagging, where each tree grows until it is complete, boosting can work with fewer splits, which makes the results much more interpretable (Hachcham 2023).

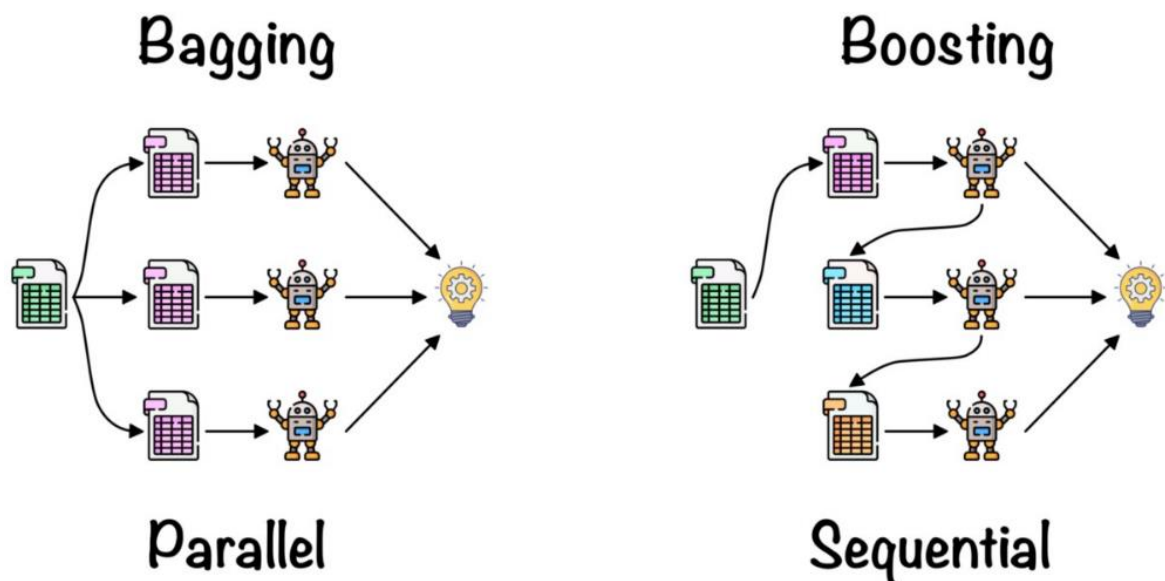


Figure 5 Parallel trees used in RF versus Sequential trees used in GBM (Source: Saini 2022)

Based on the article about the pros and cons of various ML algorithms from Shailaja Gupta (Gupta 2020) the advantages of GBM in forecasting include:

1. Simplicity of interpretation, as the data is represented in a straightforward tree structure without the need for complicated mathematical descriptions.

2. The model autonomously finds the dependencies between dependent and independent variables and selects the most significant predictors.
3. Ability to use variables with missing values.
4. Possibility to use both quantitative and qualitative variables.
5. Possibility to capture complex interactions in the dataset.

1.3.3 Neural Networks

A neural network is an ML model that emulates the way the human brain processes information. This approach consists of interconnected nodes, similar to neurons, to build the connection between two or more variables by learning from a set of patterns (Gurney 1997).

Recurrent Neural Networks (RNNs)

Recurrent neural network (RNN) is a type of neural network where the data is sequential, and each step is treated as input for the next step. Unlike traditional neural networks all the inputs and outputs are interconnected. RNNs have a “memory”, which is used to capture information about the calculations made so far (Singh 2021).

Long Short-Term Memory Networks (LSTMs)

One example of RNN is LSTM. It is built to address the issue of RNN’s failure to learn in the presence of past observations greater than 5–10 discrete time steps between relevant input events and target signals. LSTM does it by introducing a memory unit called cell state.

There are 3 layers in the cell state: the forget layer, the input layer, and the output layer. Forget layer removes irrelevant information from the previous cell state based on the current input. The input layer works in a similar way where it removes any unnecessary information from the current input and saves the important information to the cell state. The output layer takes the current input and current cell state, then outputs the hidden state and cell output, then it is scaled to a range from -1 to 1 (Singh 2020).

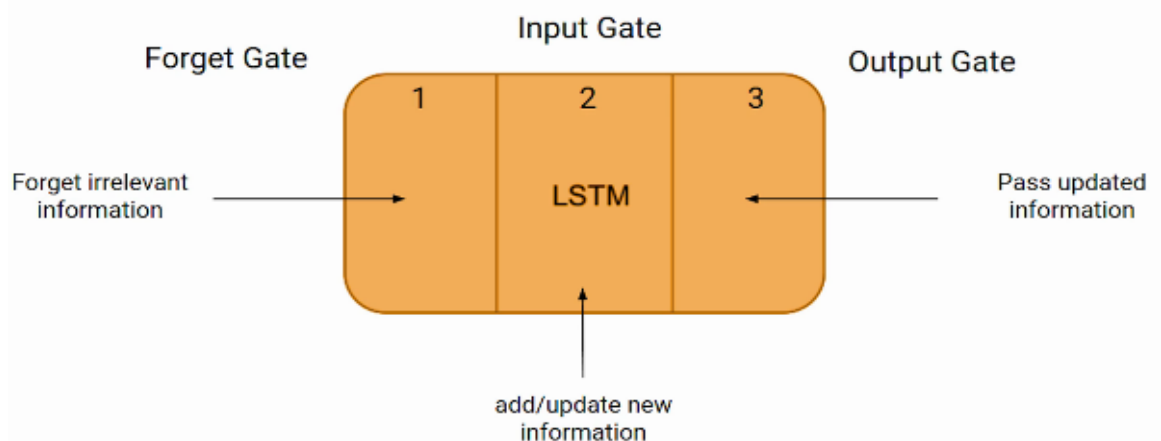


Figure 6 LSTM Cell Structure (Source: Mamun 2023)

The advantages and challenges associated with Long Short-Term Memory (LSTM) networks are numerous, with some of the most significant points highlighted by Prudhviraju Srivatsavaya in a Medium post titled "LSTM — Implementation, Advantages and Disadvantages" (Srivatsavaya 2023):

Advantages

- **Handling long sequences:** LSTMs are particularly good at working with long sequences of data. They can remember important information for longer periods, which is beneficial for analyzing time series data where past information is significant.
- **Dealing with different sequence lengths:** LSTMs can process input data of varying lengths, which is often the case in real-world situations. They adjust their memory to accommodate the length of each sequence, which is essential when the amount of input data is not consistent.
- **Avoiding vanishing gradient problem:** One of the key strengths of LSTMs is their ability to avoid the vanishing gradient problem. This problem can make it hard to train neural networks as it causes the information and gradients to shrink and become ineffective over time. The gating mechanisms in LSTMs, like the forget gate, help maintain the strength of the gradients, allowing for effective training over many time steps.

Challenges and considerations

- **Computational complexity:** LSTMs typically demand higher computational resources compared to simpler neural networks, like feedforward networks or basic RNNs. The sophisticated nature of LSTMs means that their training process can be more resource-intensive and may take longer, particularly when the network architecture becomes complex.
- **Risk of overfitting:** As with many deep learning architectures, LSTMs can be overfit to the training data, especially when the amount of data is limited. This issue can be addressed through regularization techniques such as dropout, which can help to generalize the model better by preventing it from becoming too focused on the training data.
- **Hyperparameter tuning:** The performance of LSTMs is heavily dependent on the tuning of hyperparameters, which include the number of units in each LSTM layer, the learning rate, and the length of input sequences. The process of tuning these hyperparameters is often exhaustive and requires a significant amount of experimentation to find the optimal combination.
- **Interpretability issues:** LSTMs, due to their black box nature, pose challenges in interpretability. Understanding how LSTMs make predictions can be difficult, which is a notable disadvantage in scenarios where the reasoning behind decisions needs to be clear and explainable.
- **Long training times:** Deep LSTM networks, particularly when trained on extensive datasets, can lead to prolonged training times. Such computational workloads often necessitate the use of advanced hardware infrastructure, like

Graphics Processing Units (GPUs) or Tensor Processing Units (TPUs), to accelerate the training process.

1.4 Traditional methods versus machine learning approaches

Power demand forecasting has traditionally relied on statistical techniques like ARIMA¹ and Exponential Smoothing, which excel at interpreting trends and seasonality within structured datasets. As the complexity of energy systems grows and the volume of data increases, ML methods are emerging as powerful tools, capable of identifying complex patterns from large datasets.

Unlike traditional methods that model relationships explicitly, ML algorithms learn directly from data, offering a more flexible approach to forecasting. However, they require more computational power and often lack the transparency of statistical models.

In this section traditional forecasting methods will be briefly compared with ML techniques, highlighting their practical applications in the energy sector.

1.4.1 Model complexity and data handling

Traditional statistical methods are relatively straightforward, making them accessible to practitioners and easier to implement. They perform well with smaller datasets and clear seasonal patterns but struggle with non-linearities and high-dimensional feature spaces. On the other hand, ML models excel in these challenging data scenarios but bring increased model complexity and the need for extensive computational resources.

1.4.2 Forecasting accuracy

Accuracy is a key differentiator. ML techniques often outperform traditional methods, particularly when the data exhibit complex behaviours and relationships that linear models cannot capture. The ensemble ML methods, like XGBoost, aggregate predictions from multiple models to improve accuracy and robustness, a strategy not inherent in traditional statistical forecasting.

1.4.3 Transparency and interpretability

Transparency and interpretability are critical in applications where understanding the rationale behind predictions is necessary. Traditional models score high on this front, offering clear insights into how input variables affect forecasts. In contrast, many ML

¹ Autoregressive integrated moving average – is a statistical technique suited for time-series data. It is based on the concept, where data at one point of time influence the data in the future of this point.

models, especially deep learning architectures, are seen as black boxes due to their non-transparent nature. However, efforts like feature importance are bridging this gap.

1.4.4 Flexibility and adaptability

Machine learning models are inherently more flexible. They can automatically detect and adapt to changes in data patterns over time, a feature not typically found in traditional forecasting models. This adaptability is particularly useful in power demand forecasting, where consumption patterns can shift rapidly due to technological, economic, and social changes.

1.4.5 Scalability

ML approaches are scalable with the increase in computational capabilities and data availability. They are designed to continuously improve as more data becomes available, unlike traditional methods that may not benefit from additional data to the same extent.

1.4.6 Integration and hybrid models

The comparative analysis would be incomplete without mentioning the trend towards hybrid models that combine the clear interpretability of traditional methods with the sophisticated pattern recognition capabilities of ML. These models aim to provide a balance between accuracy and understandability, potentially offering the best of both worlds for power demand forecasting.

1.5 Evaluation metrics

In the domain of forecasting, the accuracy and reliability of predictive models are critical for the effective planning and operation of the power grid. Various evaluation metrics are used to calculate the performance of machine learning models, each highlighting different aspects of accuracy and error characteristics.

This section outlines the most used metrics in power demand forecasting. The following explanation for performance metrics is as follows by Aayush Bajaj (Bajaj 2023).

1.5.1 Mean Absolute Error (MAE)

Mean Absolute Error is the average difference between the desired solutions and predicted solutions. A lower MAE indicates better model performance, with 0 representing a perfect forecast. It is described as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

where y_i is the predicted value of instance i , $\hat{y}_i$ is a real target value of instance i , and n is the total number of instances.

This measure does not put as much weight on outliers since the errors are measured linearly. This helps to get a more even measure of performance, but it means that large and small errors are weighted the same (Burch 2023).

1.5.2 Mean Squared Error (MSE)

Mean Squared Error measures the difference between the predicted solutions and desired solutions. The smaller the MSE, the better the results are. The formula is:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

where y_i is the predicted value of instance i , $\hat{y}_i$ is real target value of instance i and n is the total number of instances.

A lower MSE indicates a model that generally forecasts more accurately, but because it squares the errors, it might disproportionately reflect the impact of large errors.

1.5.3 Root Mean Squared Error (RMSE)

Root Mean Squared Error is the square root of MSE and serves to scale the error to the same units as the forecasted values. It is given by:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

where y_i is the predicted value of instance i , $\hat{y}_i$ is a real target value of instance i and n is the total number of instances.

RMSE is particularly useful when you want to understand the error magnitude in its original units. Like MSE, it gives a higher penalty for large errors.

1.5.4 Mean Absolute Percentage Error (MAPE)

The Mean Absolute Percentage Error expresses accuracy as a percentage of the error and is calculated as:

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (6)$$

MAPE is easy to interpret but can be misleading if dealing with zero or near-zero actual values. It is especially useful when you want to convey the accuracy of forecasts in a percentage, which can be more intuitive.

1.5.5 Coefficient of Determination (R^2)

The Coefficient of Determination, commonly denoted as R^2 , measures the proportion of the variance in the dependent variable that is predictable from the independent variables. It provides a sense of how well unseen samples are likely to be predicted by the model.

R^2 always takes on a value between 0 and 1. The closer R^2 is to 1, the better the estimated regression equation fits or explains the relationship between X and Y.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

where n is the number of observations, y_i is the actual value of the i -th observation, $\hat{y}_i$ is the predicted value for the i -th observation, and $\bar{y}$ is the mean of the actual values.

2 Practical part

This chapter concentrates on the practical application of the ML methods discussed in the previous section, aiming to forecast power consumption in Germany. The chapter starts by listing the specific tools and technologies used for the implementation. Then followed by an explanation of the data collection process and the various steps taken to ensure the data quality, such as cleaning and normalization, making it suitable for analysis.

Next, the forecasting models will be presented, mentioning the underlying assumptions, and detailing the preprocessing steps needed for each specific case. After this, the performance of these methods will be evaluated, and a comparative analysis will be made to determine the most effective approaches in predicting power demand.

The source code is available in Appendix A.

2.1 Tools and technologies

This thesis uses a set of tools and technologies, each selected for its specific capabilities in handling, analyzing, and modeling time-series data. Below is an overview of the key tools used. All the functionalities implemented in the practical part of this BT were written in Python programming language of version 3.11. Below is a list of the libraries that were used during the implementation.

Pandas

Pandas is the most popular data manipulation library built on top of Python. The reason for its popularity is speed, flexibility, and easiness of use. It efficiently handles and transforms the dataset into a suitable format, with functions that make operations like data filtering, grouping, and aggregation easy to do.

NumPy

NumPy is another core library for numerical computing in Python, that provides support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. It is particularly useful for performing high-level mathematical functions, such as array creation and manipulation, statistical computations like calculating means or medians, and basic arithmetic operations like squaring, efficiently on large datasets without the need for explicit loops.

Matplotlib

Matplotlib is a plotting library for the Python programming language. It provides an object-oriented API for embedding plots into applications. In this thesis it is used to visualize data

and results, such as the relationship between features and power load, and the diagnostics of the models.

Seaborn

Seaborn is a statistical data visualization library built on top of Matplotlib. It offers a higher-level interface for drawing statistical graphics. In this thesis, Seaborn is used to create more complex plots, such as heatmaps for correlation matrices.

Scikit-Learn

Scikit-Learn is one of the most popular ML libraries for Python. It features various classification, regression, and clustering algorithms, including support vector machines, random forests, gradient boosting, and k-means, but the functionalities we are interested in the most are XGBoost and evaluation metrics, such as R^2 . It is the primary toolkit used to apply ML models and additional functions associated with it, like scaling in MinMaxScaler².

Statsmodels

Statsmodels is a Python module that allows exploring data, estimating statistical models, and performing statistical tests. It complements Scikit-Learn by providing more detailed statistical methods, which are essential for in-depth time-series analysis. In this thesis, Statsmodels is particularly useful for its time-series analysis tools and for the implementation of statistical tests to verify the assumptions of linear regression. It is also used for the implementation of LR, as it allows for a formula to be written manually, to use categorical variables, and to print a summary with all the specifications of the model.

2.2 Data collecting

The weather dataset used in this thesis was taken from the OpenWeather API. The meteorological data for Germany were obtained through the “historical bulk” option offered by OpenWeather API, which allows the retrieval of past weather conditions over a specified period.

The period covered by this dataset spans from 1st January 2009 to 27th January 2024. For this thesis the data would only be considered from 30th September 2018. Before this date, Germany's power load data also included Austria, as they were part of the same bidding zone. Post this date, Germany's power load data is reported alongside Luxembourg,

² MinMaxScaler is a normalization technique that scales all values of each feature in a dataset to a specified range, typically from 0 to 1. In this scaling process, the maximum value of each feature is transformed to 1, and the minimum value to 0. It is useful when there is a need to preserve the ratio between the values (Banerjee 2023).

marking a distinct bidding zone that excludes Austria. This change is crucial for the accuracy and relevance of our analysis.

The dataset containing Germany's power load was sourced from the ENTSO-E Transparency Platform. This data was gathered in annual increments and then combined into a single data frame.

The column names in the dataset were changed for easier understanding and data handling. The data was then resampled to change the original quarter-hourly records into hourly data points using the mean function. This step helped standardize the frequency of the data.

After resampling, any missing values, shown as NaN in the dataset, were found and removed. Cleaning the dataset this way was important to ensure the later analyses were accurate and reliable.

2.3 Data exploration

This chapter goes into more detail about the variables influencing power demand, including temperature, wind, humidity, and weather. It will investigate these factors using various graphical representations, such as time series line plots, histograms, and heatmaps. The chapter will also discuss the reasoning behind the aggregations applied to each variable and their necessity for this BT. Additionally, explanations will be provided on the importance of each variable in the context of power demand forecasting.

2.3.1 Temperature

Temperature plays an important role in the dynamics of power demand. Residential and commercial heating and cooling needs are directly tied to the external temperature, creating a connection between temperature fluctuations and power usage patterns. As temperatures deviate from a comfortable "neutral" zone (typically between 18°C to 24°C), the demand for heating or cooling systems increases, subsequently elevating the power demand.

The dataset includes hourly temperature data, providing a granular view of the weather conditions and their impact on power consumption. This high-resolution data allows for the analysis of daily temperature cycles and their correlation with peak power demand times. For instance, cooler temperatures in the early morning and late evening can lead to spikes in heating demand during winter, while hotter afternoons can cause increased use of air conditioning units in summer.

Originally reported in Kelvins, the temperature data were converted to degrees Celsius for practical understanding and analysis. This conversion is crucial as air temperature is a primary factor affecting power demand.

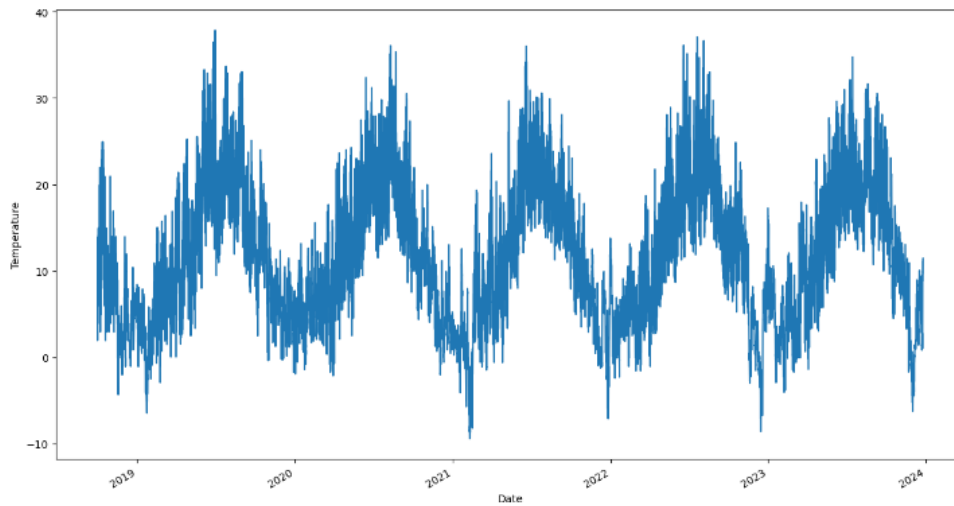


Figure 7 Daily temperature average line graph (Source: author)

The time series plot visually represents temperature trends from September 2018 through 2024. The data presents a clear seasonal pattern, with temperature peaks occurring in the summer months and troughs in the winter months each year. The amplitude of these fluctuations suggests significant variance in temperatures, with highs potentially reaching above 35°C and lows dipping below -5°C. The graph also indicates some interannual variability, as the peaks and dips are not uniform across years, which might be indicative of underlying climate patterns or anomalies.

The heatmap provides a bivariate distribution of temperature across different hours of the day for each month. The horizontal axis represents the month of the year, while the vertical axis shows the hours of the day. The color gradient from blue to red indicates the transition from cooler to warmer temperatures.

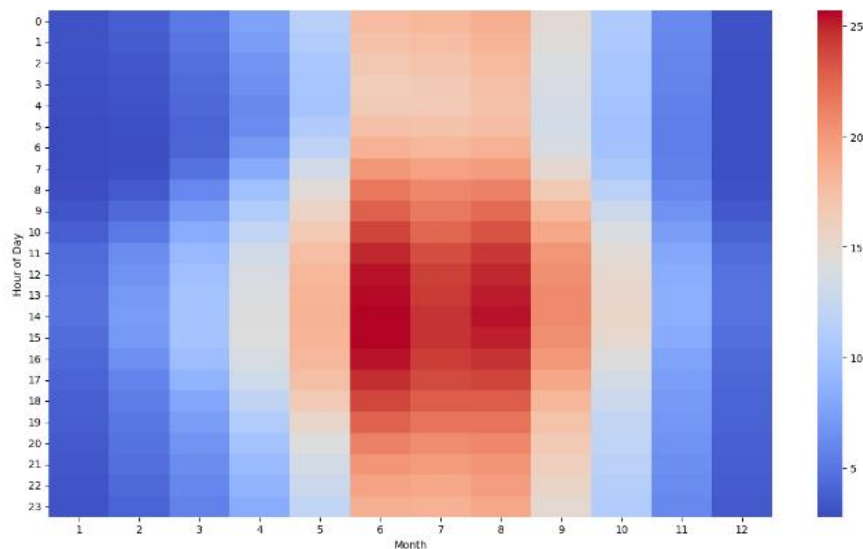


Figure 8 Temperature heatmap (Source: author)

From this visualization, we can see that the warmest part of the day is typically in the afternoon hours, particularly from June to August, which is highlighted in red. Conversely,

the cooler temperatures are mostly seen during the early morning hours across all months, as indicated by the blue shades.

2.3.2 Wind speed

Wind speed is a significant meteorological factor influencing power demand, particularly due to its impact on heating requirements and the potential for wind power generation. High wind speeds can lead to increased heat loss from buildings, thereby increasing the demand for heating during colder periods. Conversely, in regions with substantial investments in wind power, increased wind speeds can lead to higher wind energy production, potentially reducing the demand for traditional power plants.

The inclusion of wind speed data allows for a better understanding of the power demand. By analyzing wind speed together with temperature data, we can identify periods where the cooling effect of wind might reduce or increase heating needs.

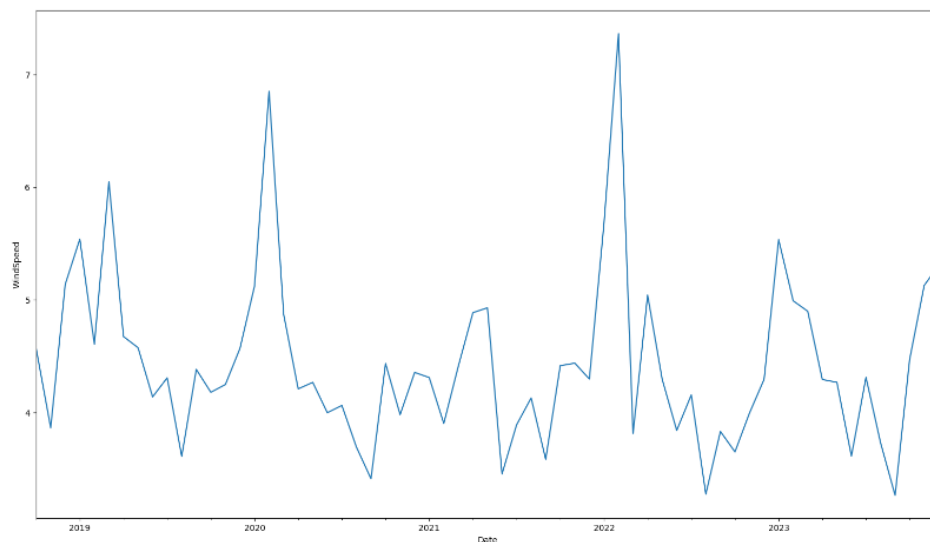


Figure 9 Monthly wind speed average line graph (Source: author)

The time series graph of wind speed displays a clear variability throughout the period from September 2018 to 2024, with no recognizable seasonal pattern like that observed in the temperature data. The wind speed peaks unpredictably, with the most notable spikes reaching above 7 meters per second.

The graph also shows periods of relative calm where wind speeds are consistently lower. This variability is critical for power systems that incorporate wind energy, as it affects the reliability and predictability of wind as an energy source.

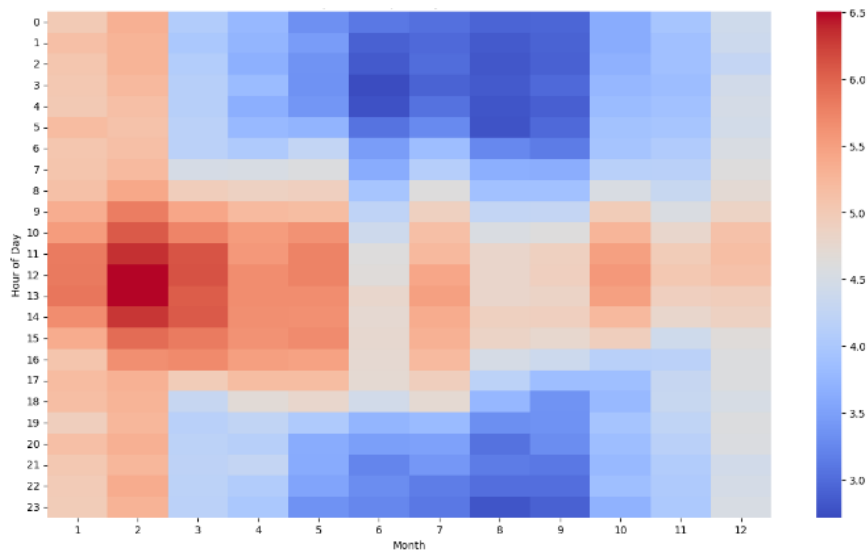


Figure 10 Wind speed heatmap (Source: author)

The heatmap offers a detailed view of how wind speed varies by hour and by month. Each cell represents the average wind speed for a given hour and month, with warmer colors indicating higher wind speeds and cooler colors showing lower wind speeds.

The heatmap shows that certain months show higher wind speeds on average, such as in the transition seasons (possibly spring and autumn). There is also a notable pattern during certain hours of the day, particularly in the late evening and early morning hours, where wind speed tends to be higher. Conversely, midday hours tend to show lower wind speeds across several months.

2.3.3 Humidity

Relative humidity is a measure of the moisture content in the air as a percentage of the amount of moisture the air can hold at a given temperature. It is an important factor in power demand analysis for several reasons:

1. **Comfort levels:** Relative humidity affects human comfort levels. High humidity can make temperatures feel warmer than they are, which can increase the demand for air conditioning in the summer. Conversely, low humidity can make the air feel cooler, potentially reducing heating demand in the winter (Chandler 2023).
2. **Heating and cooling efficiency:** Humidity levels can influence the efficiency of heating and cooling systems. Air conditioning units, for example, must work harder to cool and dehumidify the air when humidity is high, leading to increased power consumption .

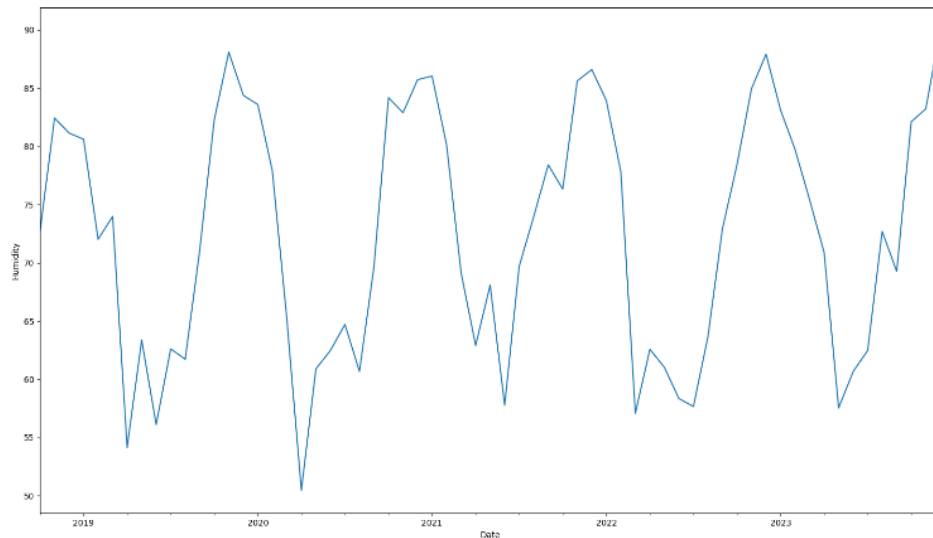


Figure 11 Monthly humidity average line graph (Source: author)

The time series plot for humidity shows a consistent range of values with some variability over the years from September 2018 to 2024. Relative humidity appears to stay mostly between 40% and 100%, with occasional dips below 40%. A clear seasonal trend is visible in the plot, which can help in the following analysis. Humidity levels tend to rise during winter months and fall in the summer, with normal winter values around 85% and in the summer around 60%.

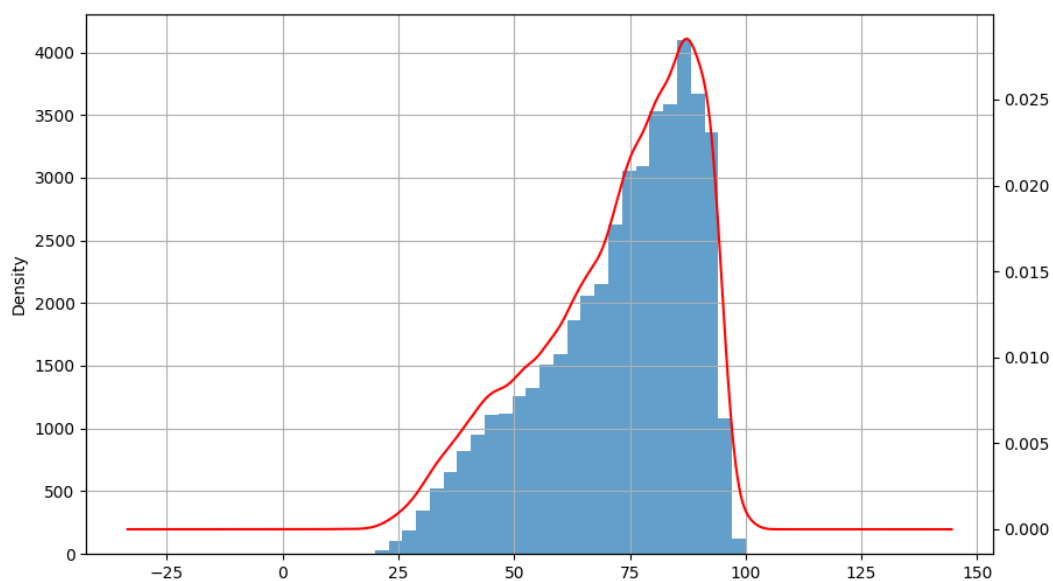


Figure 12 Histogram and Density plot of Humidity (Source: author)

The histogram, coupled with the density plot, shows the distribution of humidity readings across the dataset. The histogram shows that most humidity values are concentrated around the 60% to 80% range, suggesting that this is the most common relative humidity range in the data. The bell-shaped density curve centered around this range indicates a normal distribution of humidity levels with a skew towards higher humidity.

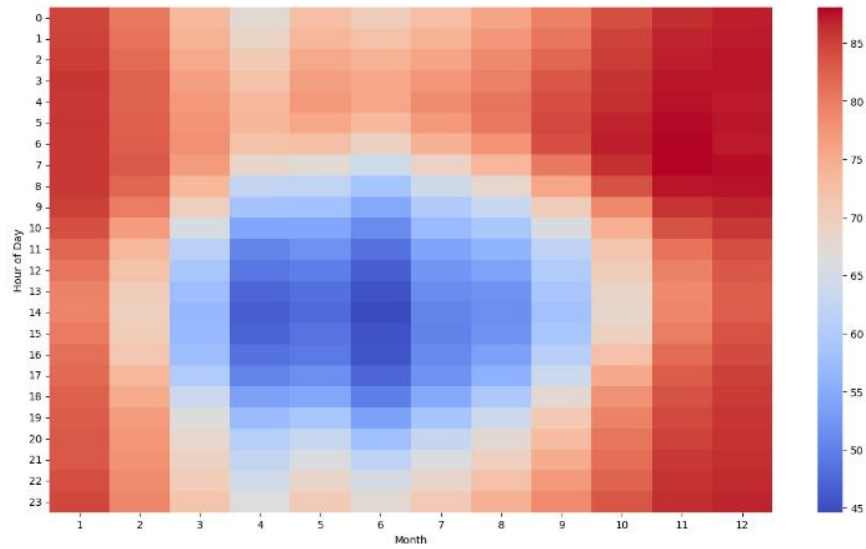


Figure 13 Humidity heatmap (Source: author)

The heatmap presents the average humidity by hour and month, with red indicating higher humidity levels and blue indicating lower levels. There is a clear daily pattern, with higher humidity typically occurring in the early hours of the morning and lower humidity in the afternoon, which is typically due to the rise in temperature throughout the day causing the relative humidity to drop. Seasonal variations are quite distinct in this visualization since there appears to be a trend towards higher humidity in the latter months of the year, which is the reflection of seasonal weather changes.

2.3.4 Weather

This variable describes general weather conditions as a categorical variable (such as clear, cloudy, rain, etc.). Categorizing weather patterns allows for a structured analysis of how different conditions impact power demand. For instance, clear days might lead to higher solar energy production, reducing reliance on conventional power sources. Cloudy or rainy days, however, could increase the demand for artificial lighting and heating.

2.3.5 Power load

The power load data, serving as the target variable in this thesis, were sourced from the ENTSO-E platform, a key provider of energy-related data in Europe. This dataset contains hourly records of power demand in Germany, denoted as "DE_Load" in the analysis.

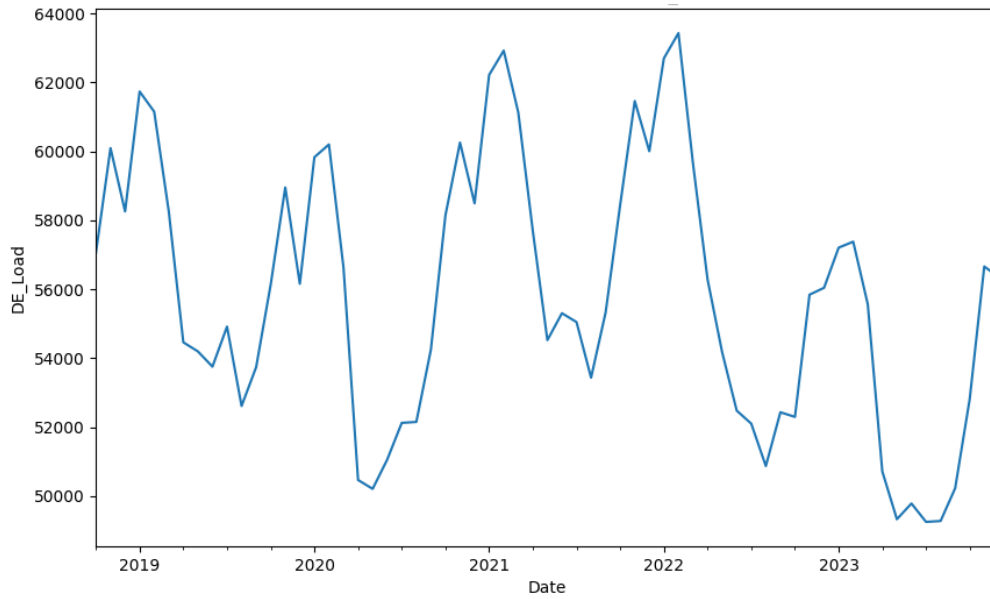


Figure 14 Average monthly power demand line graph (Source: author)

The plot shows periodic changes in power demand, which correspond to seasonal variations. Higher peaks are typically seen in winter, showing increased energy usage due to heating requirements. The lower points in the plot are indicative of the summer or transitional seasons where the demand for heating is lower. This seasonality is a common pattern in power consumption due to residential and commercial buildings' heating and cooling needs.

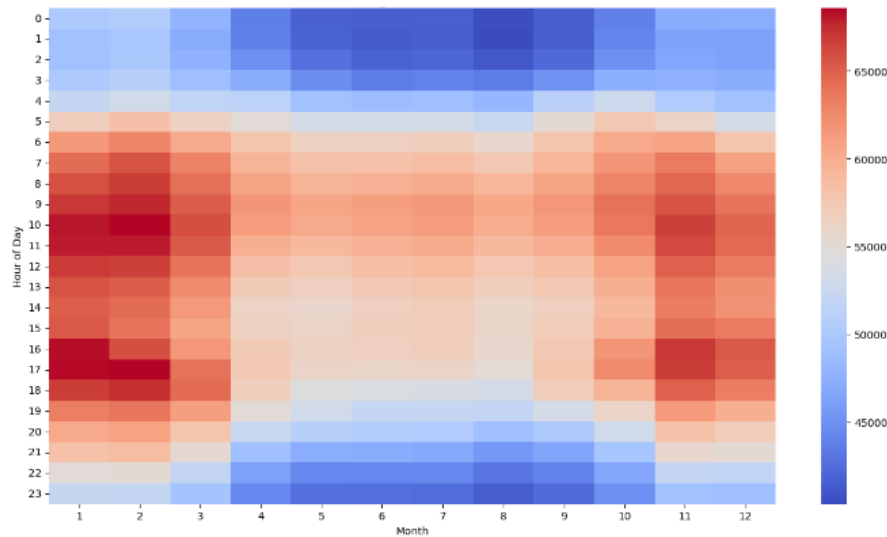


Figure 15 Power demand heatmap (Source: author)

This heatmap provides a visual representation of the power demand in Germany, segmented by hour of the day and by month. The color gradient, ranging from blue to red, represents the magnitude of the power load, with blue indicating lower demand and red indicating higher demand.

The horizontal axis of the heatmap is divided into months of the year, from January to December, while the vertical axis is segmented into hours of the day, from 0 to 23. Each cell in the heatmap corresponds to an average power load value for a specific hour and month.

The red tones, which show higher power demand, are more visible in certain months, mostly during winter. These periods of high demand could correlate with cold winter months, when heating requirements increase, and during working hours when residential and commercial activities peak.

On the other side, the blue tones, indicating lower power demand, are noticeable during the warmer months and possibly during the midday hours when temperatures are higher, or during the night when there is less activity.

2.4 Data preprocessing

The preprocessing of the dataset involved the normalization of numerical data to ensure consistency and comparability across measurements. The “Temperature” variable underwent a conversion from Kelvin to Celsius scale by subtracting 273.15 from each temperature value. Additionally, the categorical variable “Weather Type” was encoded into two categories: “Good” and “Bad” weather conditions. “Good” was defined to include conditions such as “Clear” and “Clouds”, while “Bad” encompassed all other weather types. The dataset was found to be free of missing data and was uniformly converted to the UTC time zone for standardization.

Subsequently, the dataset was partitioned into training and testing sets following an 80/20 ratio. The training set is designated for identifying patterns and dependencies, whereas the testing set serves as a validation and performance evaluation tool.

In the preparation for LR, it was also necessary to encode time-related variables. Specifically, categorical variables representing the month, day of the week, and hour were derived from the timestamp data. Since the data is cyclical, this transformation allows for the inclusion of seasonal effects in the analysis, providing a better understanding of how these time dimensions influence the target variable.

Then a categorical variable, designated as “War” was introduced into the dataset to encapsulate the geopolitical factor pertaining to the escalation of the conflict in Ukraine on 24th February 2022. This variable, defined as a Boolean type, assumes a value of “True” after the mentioned date, thereby providing a binary indication of the pre-war and post-war periods within the analytical framework.

2.5 Solutions

In this section, the main focus will be on describing the implementation steps of the models. It will cover multiple ML models, particularly LR, XGBoost, and LSTM. It will also examine if the discussed data meets the LR assumptions, investigate which features are most important, and outline the steps needed for each model. After that there will be a comparison of the results, checking how well the predictions align with actual data. Additionally, it will discuss the limitations that the approaches faced and draw the conclusion on overall efficiency in applying the mentioned models in practice.

2.5.1 Regression analysis

Linear regression assumptions

In this chapter the assumptions for LR model will be examined, which is a critical step for ensuring the validity and reliability of forecasting results. Linear regression relies on several key assumptions including linearity, independence, homoscedasticity, and normality among the residuals. A clear understanding of these assumptions helps in model specification and interpretation. The aim of checking these assumptions in the context of power demand forecasting model is to improve the approach and validate whether LR is a suitable method.

Linearity

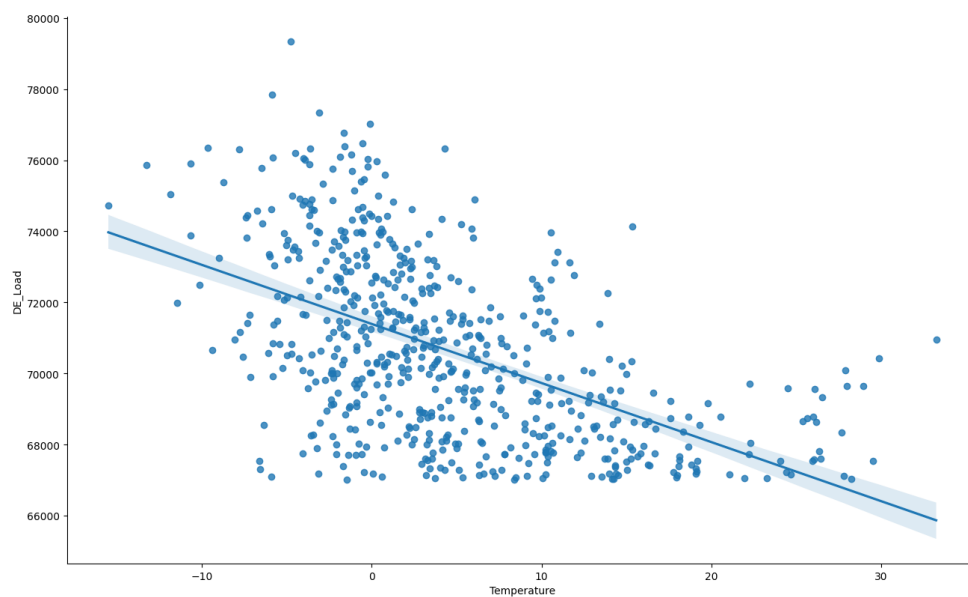


Figure 16 Linearity graph of Temperature (Source: author)

This scatterplot of Temperature against Load should ideally display a linear pattern, indicating that the load changes at a consistent rate as temperature increases or decreases. While we can see that the graph shows a downward trend, the data points are quite scattered and do not closely follow the strictly linear trend.

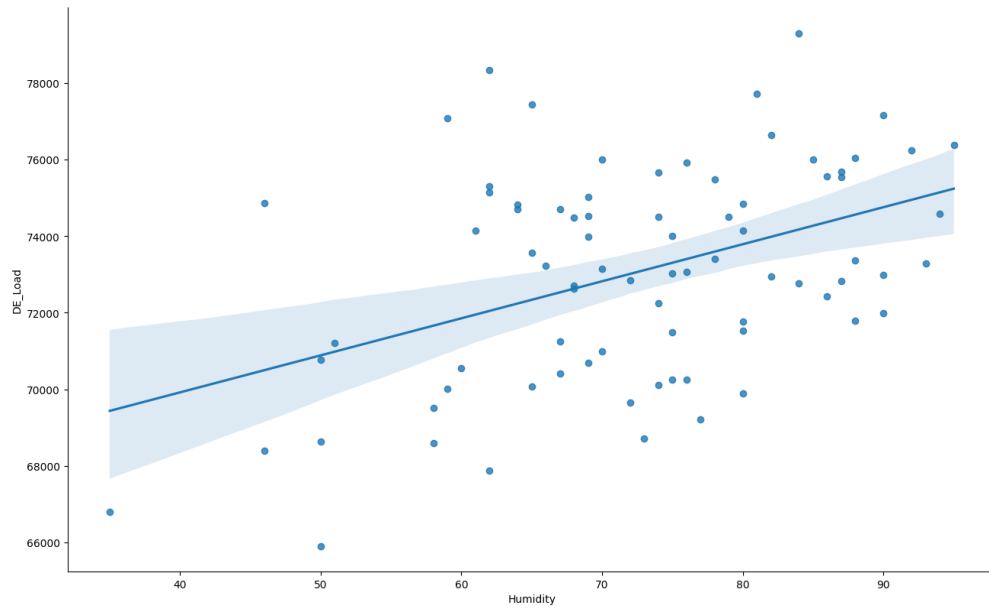


Figure 17 Linearity graph of Humidity (Source: author)

The next graph plots Humidity against Load, indicating a positive linear relationship, as suggested by the upward trend line. However, data points seem to stray from the general trend, suggesting high variability and showing that a linear model may only partially capture the relationship between humidity and demand. A residual analysis would be help in confirming the linearity assumption and to check for other potential violations, such as heteroscedasticity or non-independence of residuals.

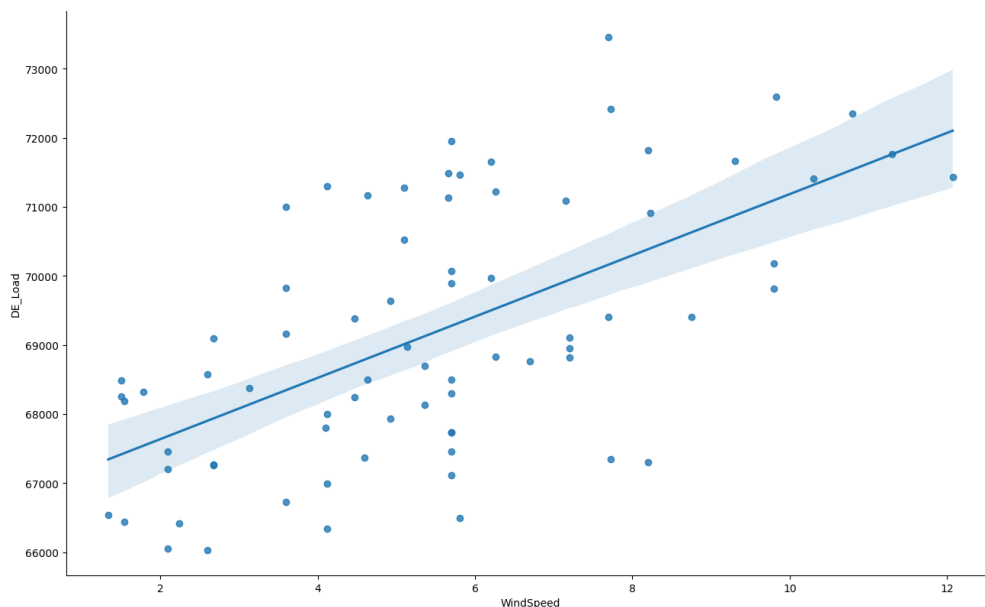


Figure 18 Linearity graph of Wind Speed (Source: author)

The scatterplot shows a generally positive linear trend, indicated by the fitted line across the wind speed values. Although the data points disperse around the line with varying

degrees of proximity, there is no immediate indication of a non-linear pattern, which confirms the linearity assumption.

Independence

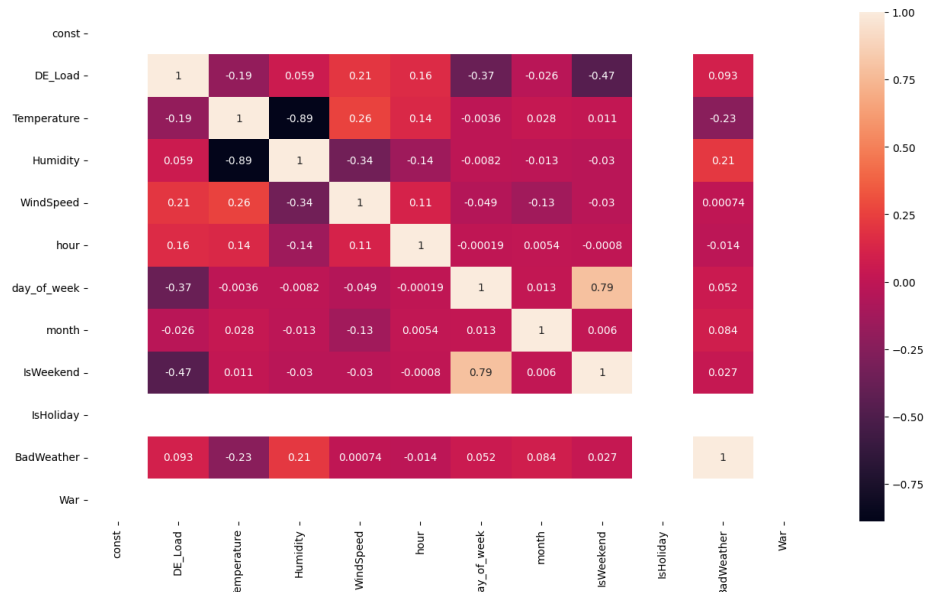


Figure 19 Variables correlation matrix (Source: author)

The graph above is a correlation matrix heatmap that visualizes the correlation coefficients between different variables. The correlation coefficient values range from -1 to 1, where 1 means a perfect positive linear relationship, -1 means a perfect negative linear relationship, and 0 implies no linear relationship.

From the heatmap, we can interpret several key points about the independence of variables:

- The diagonal, typically filled with value of 1, represents the correlation of each variable with itself, which is always perfectly correlated.
- The correlation coefficients between the power load and other variables like temperature, humidity, and wind speed are relatively low, which suggests that these predictors independently contribute to the variation in power load without being highly interdependent. This is a good indication that multicollinearity may not be a significant issue for these variables.
- There is a notably high negative correlation between temperature and humidity (-0.89). This suggests a strong linear relationship and potential multicollinearity, which could affect the model if both variables are included as predictors. In linear regression, high multicollinearity can inflate the variance of the coefficient estimates and make the model less reliable.
- Day of the week seems to have a strong negative correlation with variables “isWeekend” and “isHoliday”, which is intuitive as weekends and holidays usually fall on specific days of the week.
- Some variables like “BadWeather” and “War” show very little correlation with the load, which could imply that they do not have a strong linear relationship with the load, or that any effect they might have is not captured linearly.

Homoscedasticity

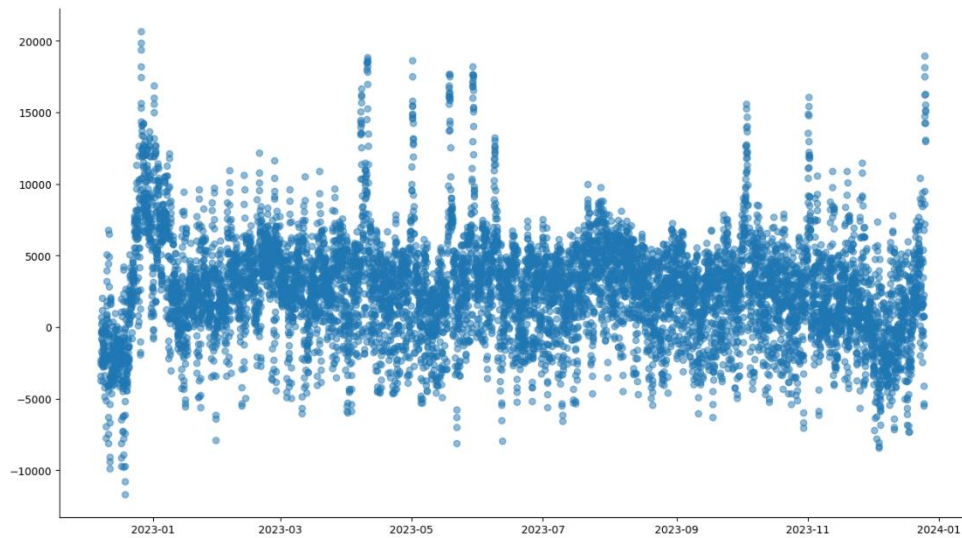


Figure 20 Residual Plot analysis for verifying the homoscedasticity (Source: author)

This graph shows the residuals from a regression model plotted over time. To assess homoscedasticity — the assumption that the residuals have constant variance across time — we would expect to see the residuals distributed randomly around the zero line, with no clear pattern or systematic change in their spread.

From the plot, it appears there may be some concerns regarding homoscedasticity. The variance of the residuals does not look consistent over time, with periods showing larger spreads than others. The presence of such clusters of high variability might suggest periods of increased volatility, which could be an indication of heteroscedasticity.

Normality

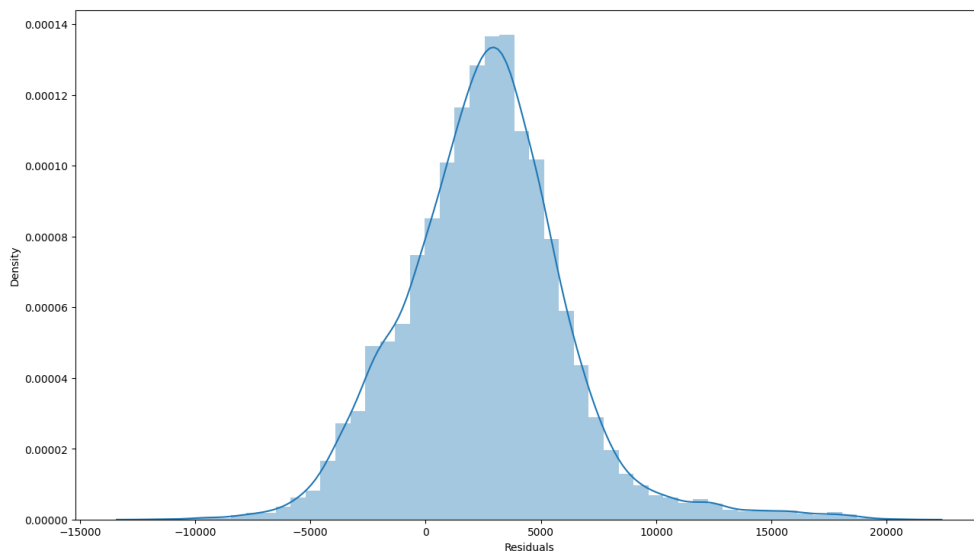


Figure 21 Distribution of residuals histogram (Source: author)

From the histogram's shape and the curve's fit, it seems that the residuals are approximately normally distributed, as the bulk of the data is centred around the mean. The symmetric bell shape is a good indicator that the normality assumption may not be severely violated.

However, it is important to note that there appears to be a slight skewness to the right, as indicated by the longer tail extending in the positive direction of the residuals. This could suggest there are larger positive than large negative errors. Also, the peak of the histogram is quite sharp, and the tails are heavier than what would be expected in a perfectly normal distribution.

Summary

OLS Regression Results			
=====			
Dep. Variable:	DE_Load	R-squared:	0.828
Model:	OLS	Adj. R-squared:	0.827
Method:	Least Squares	F-statistic:	3898.
Date:	Mon, 15 Apr 2024	Prob (F-statistic):	0.00
Time:	19:56:10	Log-Likelihood:	-3.5589e+05
No. Observations:	36566	AIC:	7.119e+05
Df Residuals:	36520	BIC:	7.123e+05
Df Model:	45		
Covariance Type:	nonrobust		

Figure 22 Statistical summary of the LR model predicting power load (Source: author)

The regression model summary provides an overview of an Ordinary Least Squares (OLS) regression analysis with "DE_Load" as the dependent variable. The model has a high R-squared value of 0.828, indicating that approximately 82.8% of the variability in the power load can be explained by the independent variables included in the model. The adjusted R-squared is similarly high at 0.827, which adjusts for the number of variables and the sample size, suggesting a good fit even after accounting for the model complexity.

The P-value is close to zero, meaning the model is statistically significant and the explanatory variables collectively have a strong linear relationship with the power load.

The formula used in this BT for the Ordinary Least Squares model is:

$$\begin{aligned}
 DE_Load \sim & Temperature + Humidity + WindSpeed + C(BadWeather) + C(hour) \\
 & + C(day_of_week) + C(month) + C(War) + \varepsilon
 \end{aligned}
 \tag{8}$$

where $C(variable)$ stands for categorical variable and ε is an error term.

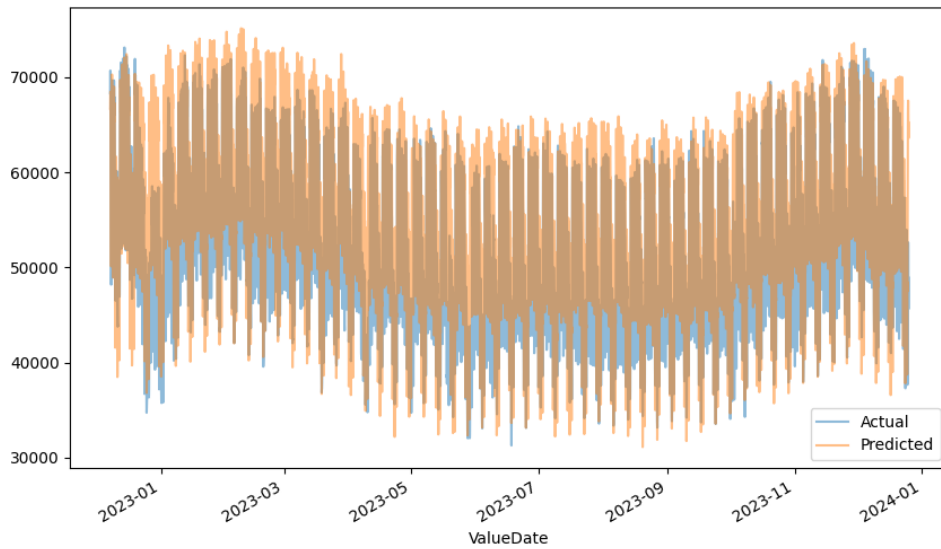


Figure 23 Actual versus Predicted values for LR (Source: author)

The graph illustrates the actual versus predicted power load values from the LR model. The blue line represents the actual load, and the orange line shows the model's predictions. While the model follows the general seasonal trend, there are noticeable inconsistencies at certain points, such as on January 1st, where the actual demand significantly drops but the prediction does not reflect this change. Despite including a variable for holidays in our model, it appears to have had limited impact on specific dates, such as January 1st, where the discrepancy between actual and predicted demand remains significant. To address this issue, we could involve postprocessing the model's outputs by applying different weights to particular holidays that have a huge impact on power demand, like New Year's Day.

Table 1 Linear regression parameters

Parameter	Coefficient
Intercept	84070
Temperature	-19790
Humidity	-31960
Wind speed	6074
Hour	11993
Day of week	-2350
Month	-3363
Year	-1197
Weather	-22

Table 1 presents the coefficients of a linear regression model used to forecast power demand. The intercept is set at 84070. Temperature and humidity are significant negative predictors, decreasing the predicted value by 19790 and 31960 respectively for each unit increase. Conversely, wind speed positively affects demand, increasing it by 6074. Hour, day of the week, and month are categorical variables that have been aggregated to an average in this table for ease of interpretation. Additionally, bad weather and the presence of war are categorical variables with minor negative impacts, decreasing the forecast by 22 and 1197 per unit change respectively.

2.5.2 XGBoost

The next modeling approach that was implemented was the Decision Tree method, particularly the XGBoost algorithm. The model showed good results having the R-squared value of 0.799, indicating a decent predictive accuracy. It is noteworthy that the XGBoost model has a more accurate prediction during the decreased demand at the beginning of the year, as can be seen in Figure 24, as compared to the LR model. This could be due to the non-linear and complex relationships that XGBoost is capable of capturing, which LR might not handle as effectively. This brings the idea of incorporation another time-based categorical variable to linear regression approach, which would indicate whether a day is a holiday.

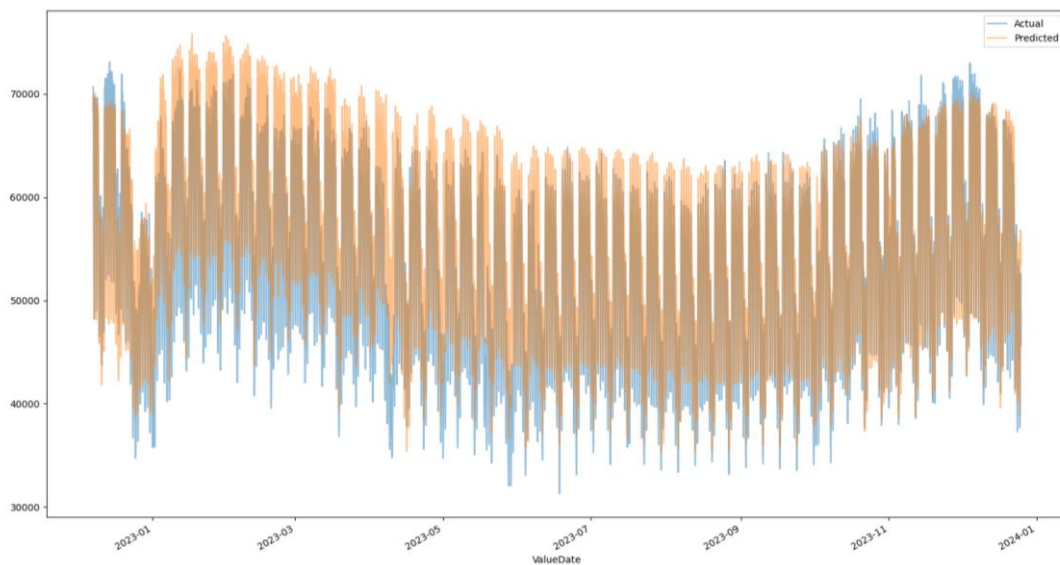


Figure 24 Actual versus Predicted values for XGBoost (Source: author)

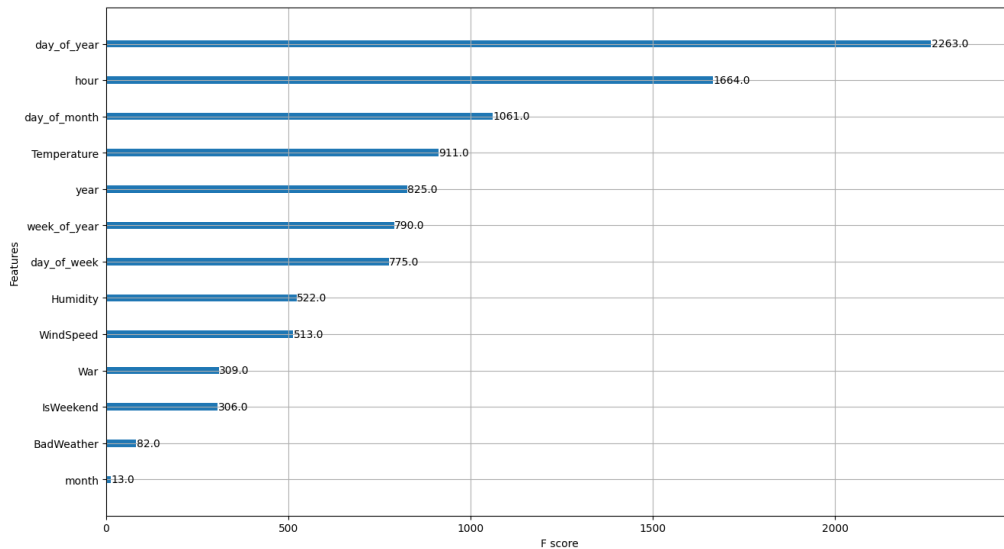


Figure 25 Ranked features importance for XGBoost (Source: author)

This bar chart illustrates the feature importance as determined by an XGBoost model applied to power demand forecasting. Feature importance is evaluated based on the F-score, which is derived from the number of times a feature is used to split the data across all trees within the model. In this chart, “day_of_year” emerges as the most significant predictor, followed by “hour” and “day_of_month”, suggesting a strong temporal component in power demand patterns. This aligns with expectations, as power usage typically fluctuates with time due to factors like natural light and human activity cycles.

The “Temperature” variable also shows substantial importance, consistent with the general understanding that temperature changes significantly impact energy consumption due to heating and cooling needs.

Surprisingly, “year” appears as a relatively important feature, indicating that year-on-year changes, potentially reflecting broader economic trends or energy efficiency improvements, influence power demand.

Other temporal features like “week_of_year” and “day_of_week” hold moderate importance, underscoring the relevance of both seasonal trends and weekly cycles, such as workdays versus weekends, in forecasting demand. The features “Humidity” and “WindSpeed” appear less critical in this model, but still relevant, suggesting that weather conditions have a non-negligible impact on power demand.

Less influential features such as “isWeekend”, “BadWeather”, and “month” appear at the bottom of the ranking.

The variable “War”, although not typical in energy forecasting models, has been included and presents a non-trivial importance score. This may reflect the model capturing specific periods of increased uncertainty or changes in energy supply and demand due to geopolitical events.

2.5.3 Long short-term memory

Preprocessing

For the purposes of this thesis, a univariate model will be implemented. This means that a single feature will be utilized to construct the model, specifically the power load in Germany. The initial step involves extracting this feature, then scaling it using the MinMaxScaler to ensure the data is appropriately normalized for the modeling process. Then, the “look back” parameter is created and set to 25, explaining the number of previous time steps to use as input variables to predict the next time period. This parameter defines the window of past information the model will consider.

For clarity, take this sequence of data: [a, b, c, d, e]. After preprocessing the structured data will be organized into input-output pairs as follows:

- Input [a, b] predicts Output [c].
- Input [b, c] predicts Output [d].
- Input [c, d] predicts Output [e].

With the data prepared, the LSTM model architecture was defined by using the Sequential API from Keras. The model consisted of an LSTM layer with 100 neurons, followed by a dropout layer set to 20% to prevent overfitting.

Summary

The chart presents the comparison of an LSTM model versus actual data, showing a high R-squared value of 0.99. A visual comparison of the actual and predicted values shows the high percentage of overlap between those, one more time confirming the accuracy of the suggested model.

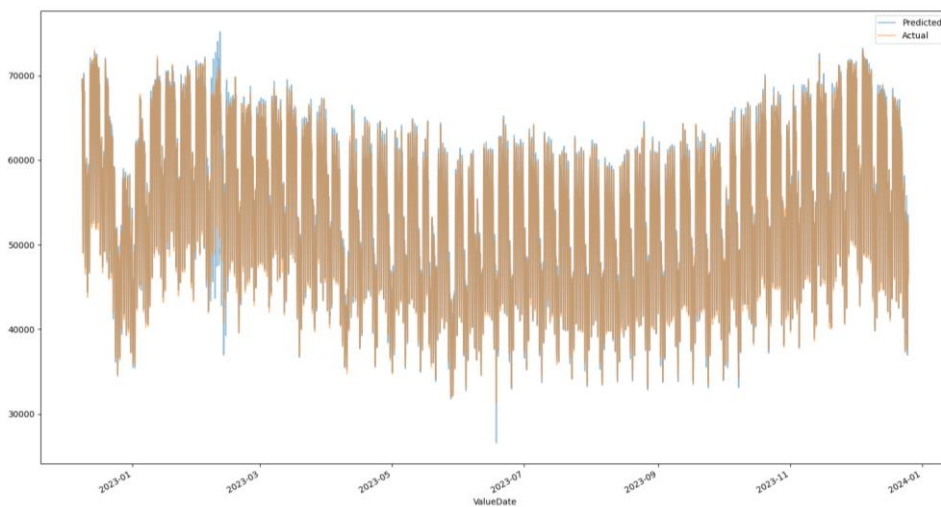


Figure 26 Actual versus Predicted values for LR (Source: author)

The LSTM model performed close to the forecasted data present in the dataset. The performance can be improved further by forming this problem as a multivariate time series and tweaking the LSTM model. There are also many parameters that can improve the overall performance.

For the construction of this model, the ADAM optimization algorithm was selected due to its reputation as an extension of stochastic gradient descent, which updates network weights in an iterative manner based on training data. The primary rationale for its selection includes its easy implementation and low memory requirements. Also, ADAM is known for its computational efficiency (Brownlee 2021).

Dropout was added as a setting in the model to help it perform better on new, unseen data. It works by randomly turning off some of the neurons in the network during training, which stops the model from relying too much on any one part of the data. This is a simple yet effective way to keep the model from overfitting, and it does not add much to the computational load. It is a widely used trick to ensure the model can handle different kinds of data well (Brownlee 2018).

Upon further investigation, it was found that when the model attempts to predict future events, it lacks the precision observed with past data. The main issue appears from the attempt to not predict future data with unknown values. The way it works during training is that it takes the last 25 values, makes a prediction for next hour, and then just shifts the window. Instead, the model should take the last 25 values to predict the next hour, and then update the windows by removing the earliest hour and adding the predicted value. This process consecutively feeds the model with its own predictions. However, during training, actual data must be provided so the neural network can learn and recognize patterns.

This approach quickly revealed a significant problem: when fed with forecast data, the predictions rapidly become inaccurate and repetitive. As illustrated in Figure 27 and Figure 28, the forecasted model output tends to repeat itself, leading to stale predictions that fail to reflect the actual downward trend in the data (Figure 28).

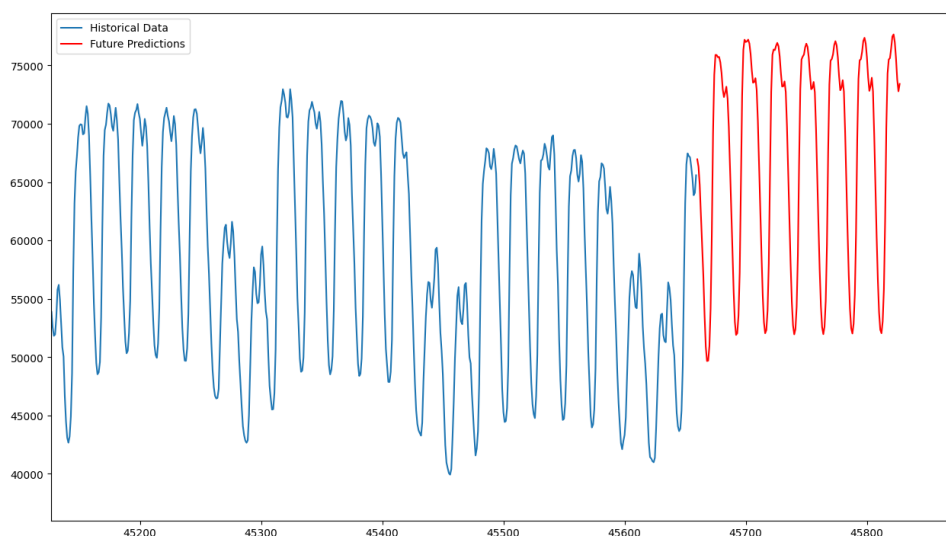


Figure 27 Initial model future prediction including historical data (Source: author)

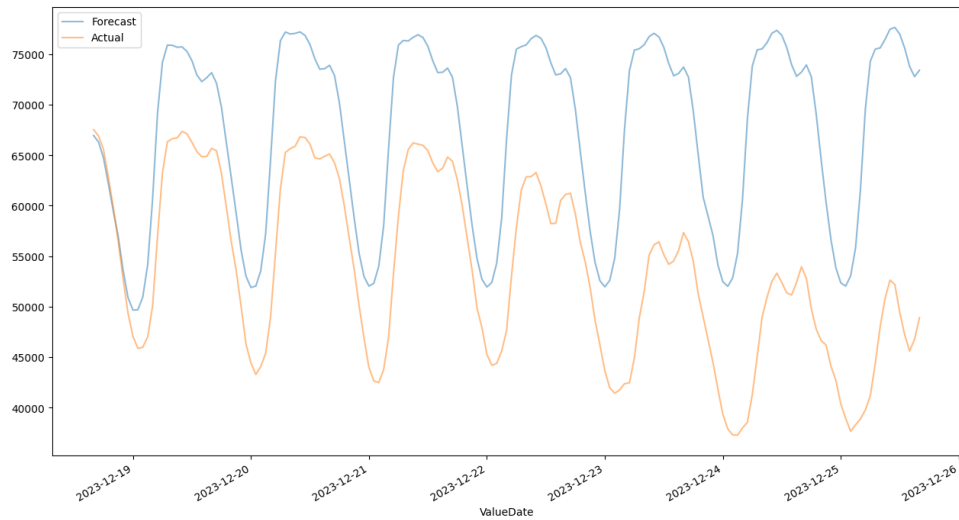


Figure 28 Initial model future prediction against actual data comparison (Source: author)

To enhance the model's performance, an alternative approach was implemented: the model's look-back period was increased from 25 to 100 to consider the last 4 days of data instead of 1. This adjustment significantly improved the model's performance, as can be seen in Figure 29 and Figure 30. Now, the model more closely follows the general trend and avoids becoming repetitive.

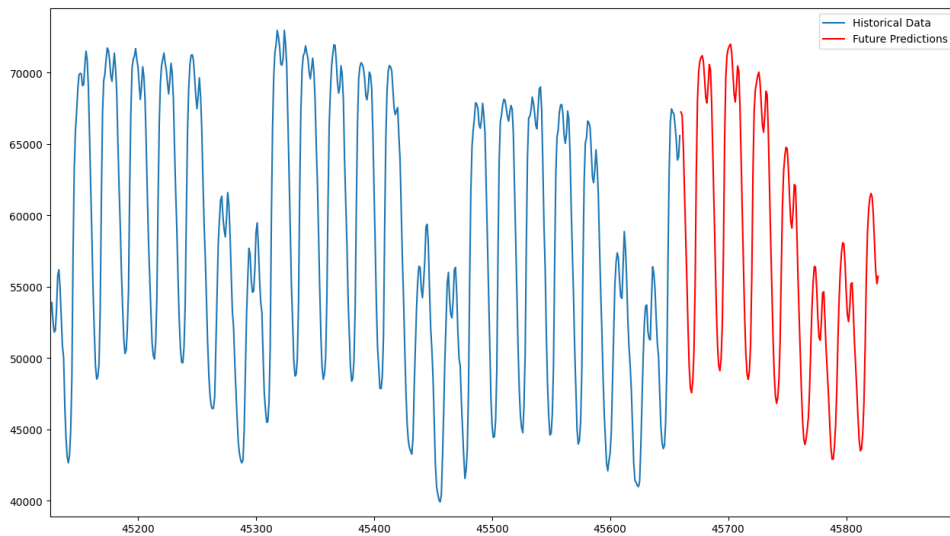


Figure 29 Model future prediction including historical data after the change (Source: author)

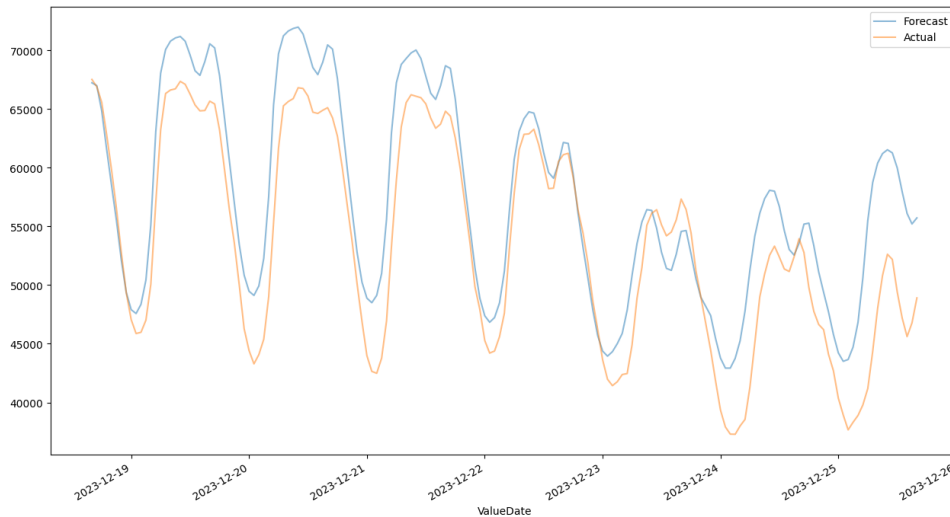


Figure 30 Model future prediction against actual data comparison after the change (Source: author)

This modification to the LSTM model has shown better results. By extending the look-back window to 100 hours, the model now demonstrates improved accuracy and overall improved outputs. This improvement suggests that the model has developed a deeper understanding of the data. These advancements provide a solid foundation for further refinement of the model and exploration of additional methods to optimize predictive performance.

2.6 Results comparison

In the results comparison chapter, the focus will be on the performance of the different models tested in this study. Presented in Table 2 are the evaluation metrics for each model—Linear Regression, XGBoost, and Long Short-Term Memory—which are fundamental to understanding their predictive capabilities. We assess the models based on MSE, RMSE, MAE, MAPE, and the coefficient of determination, R^2 , alongside the training time in seconds.

Each metric in the table offers insights into the models' performance metrics. MSE and RMSE are indicators of overall prediction error, MAE presents the average of the absolute errors, and MAPE represents error as a percentage, which is easy to interpret. The R^2 value measures the fit of the predictive model to the actual data, while training time indicates computational efficiency.

Table 2 reveals that the LSTM model outperforms the others with an R^2 value of 0.99, indicating that it predicts the actual data with 99% accuracy. However, it is important to note that this high accuracy was observed on training data; real-world predictions exhibited slightly lower precision, though they were still quite accurate. Despite its effectiveness, the LSTM model operates at a significantly slower computational speed, being 100 times slower than XGBoost. While this speed is generally adequate, transitioning to a multivariate model would substantially increase the computation time, a factor that should be considered when

choosing a model. Another disadvantage of LSTM is the lack of interpretability, which can be critical in some applications.

When comparing the two LSTM models, the initial model with a look-back window of 25 displayed better evaluation metrics, indicating more accurate predictions for the next-hour value compared to the model with the extended window. This model would be preferred for short-term forecasting, such as next-hour predictions. However, for longer-term forecasting, the second LSTM model, which has a look-back period extended to 100 hours, should be chosen as it more effectively captures and follows weekly trends.

XGBoost and Linear Regression show more comparable results. XGBoost has slightly lower error metrics, while Linear Regression's predictions fit the actual data slightly better, with a 2% higher R^2 value. The training times of Linear Regression and XGBoost are close enough to not consider—1.37 seconds for LR and 2.64 seconds for XGB. Both models are interpretable. The importance of features for XGBoost was illustrated in Figure 25. For LR the formula was provided and the coefficients for each variable are easily retrievable.

Table 2 Evaluation Metrics of different models (Source: author)

Model	MSE	RMSE	MAE	MAPE	R^2	Training time (in seconds)
LR	19514854	4417	3482	0.0708	0.82	1.37
XGB	16900557	4111	3201	0.0631	0.799	2.64
LSTM (look back = 25)	691024	831	577	0.0111	0.99	279
LSTM (look back = 100)	1141086	1068	786	0.0158	0.98	331

Conclusion

Power demand forecasting is a critical component in the management of energy resources, making sure that electricity generation aligns efficiently with consumption patterns. Accurate forecasting allows different companies working in the energy business to optimize their operations, reduce costs, and maintain grid stability.

In the theoretical part, the structure of the German and European energy grids was explained, as well as all the information that was necessary for the practical part. This included explanations of various machine learning algorithms, evaluation metrics, and their operational mechanisms.

In the practical part, this bachelor thesis has tested the application of various machine learning models to improve power demand forecasting, with a specific focus on integrating meteorological data such as temperature, wind, and humidity. The comparative analysis revealed that certain models, particularly the long short-term memory, performed better by providing high accuracy in predicting power demand. These findings highlight the potential of machine learning techniques to improve the precision and reliability of forecasts, which is critical for energy grid management and the integration of renewable energy sources.

This thesis contributes to the field of power demand forecasting by showing the practical uses and limitations of various machine learning methods. The findings emphasize the critical role of model selection in achieving a balance between accuracy, computational efficiency, and interpretability. Choosing a single model is challenging due to the distinct advantages and drawbacks of each approach. Linear regression offers the simplest interpretation, long short-term memory provides the highest performance whilst being slower, and XGBoost represents a compromise between interpretability and accuracy. Overall, this work supports the goal of increasing the efficiency and reliability of power supply systems through better demand prediction techniques, highlighting the importance of understanding both the energy infrastructure and the latest machine learning approaches.

However, the thesis also faced some challenges, such as data availability and model scalability, that could impact the broader application of the solution. Future research may focus on expanding the data sets used, exploring real-time forecasting, and testing the models in diverse environmental settings. Additionally, further investigation into hybrid models that combine the strengths of multiple machine learning techniques could provide even greater improvements in forecast accuracy.

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Appendices

Appendix A: Source code

File code.zip contains all the source code.